Extract from:

# Linear description of ultrasound imaging systems

Notes for the
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Jørgen Arendt Jensen

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Ørsted • DTU, Build. 348,
Technical University of Denmark
DK-2800 Lyngby, Denmark
E-mail: jaj@oersted.dtu.dk
Web: http://es.oersted.dtu.dk/staff/jaj/

### 3.5.1 Simulation model

The first simple treatment of ultrasound is often based on the reflection and transmission of plane waves. It is assumed that the propagating wave impinges on plane boundaries between tissues with different mean acoustic properties. Such boundaries are rarely found in the human body, and seldom show on ultrasound images. This is demonstrated by the image shown in Fig. 3.14. Here the skull of the fetus is not clearly marked. It is quite obvious that there is a clear boundary between the fetus and the surrounding amniotic fluid. The skull boundary is not visible in the image, because the angle between the beam and the boundary has a value such that the sound bounces off in another direction, and, therefore, does not reach the transducer. Despite this, the extent of the head can still be seen. This is due to the scattering of the ultrasound wave. Small changes in density, compressibility, and absorption give rise to a scattered wave radiating in all directions. The backscattered field is received by the transducer and displayed on the screen. One might well argue that scattering is what makes ultrasound images useful for diagnostic purposes, and it is, as will be seen later, the physical phenomena that makes detection of blood velocities possible. Ultrasound scanners are, in fact, optimized to show the backscattered signal, which is considerably weaker than that found from reflecting boundaries. Such reflections will usually be displayed as bright white on the screen, and can potentially saturate the receiving circuits in the scanner. An example can be seen at the neck of the fetus, where a structure is perpendicular to the beam. This strong reflection saturates the input amplifier of this scanner. Typical boundary reflections are encountered from the diaphragm, blood vessel walls, and organ boundaries.

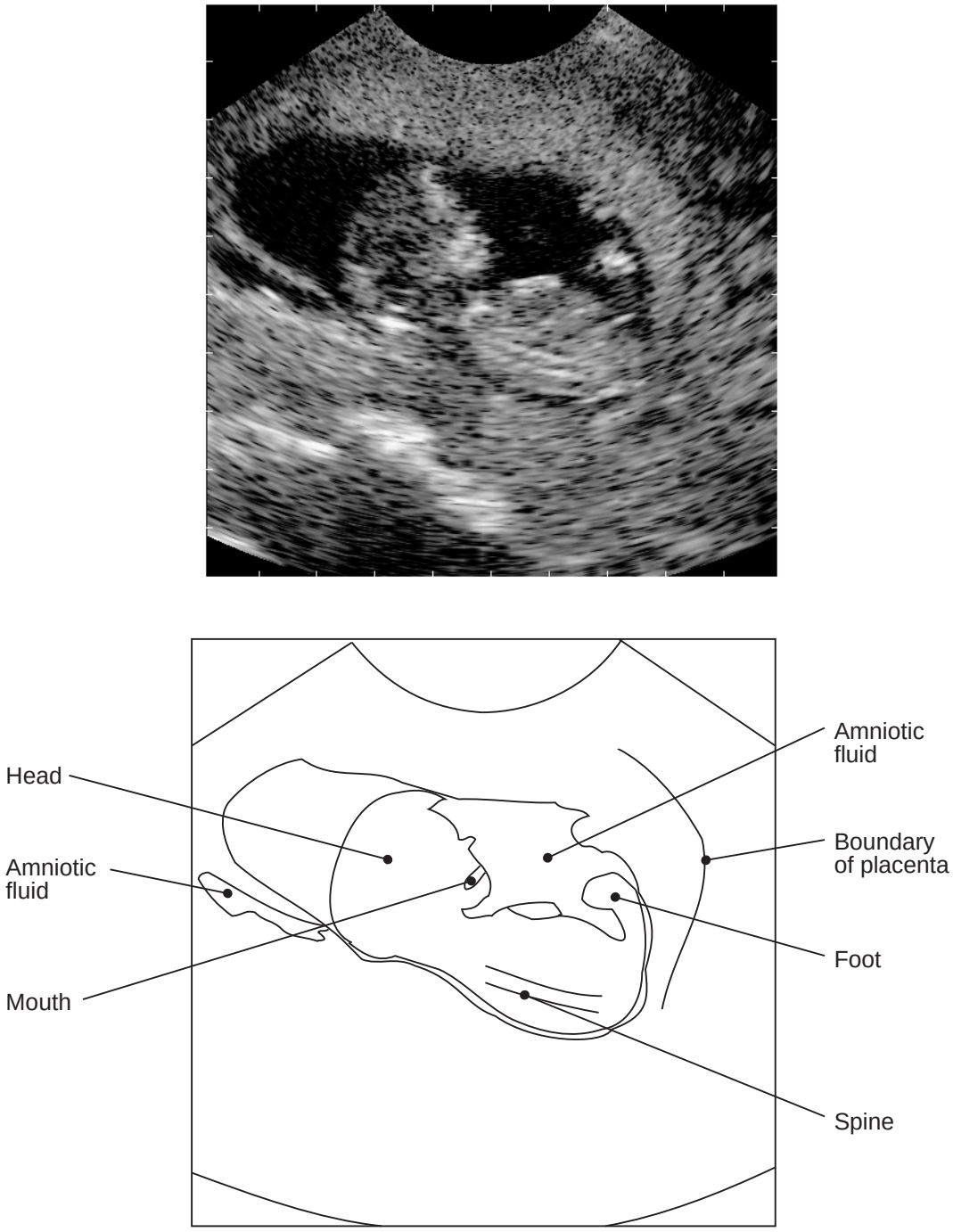



Figure 3.14: Ultrasound image of a 13th week fetus. The markers at the border of the image indicate one centimeter (from [4], Copyright Cambridge University Press).

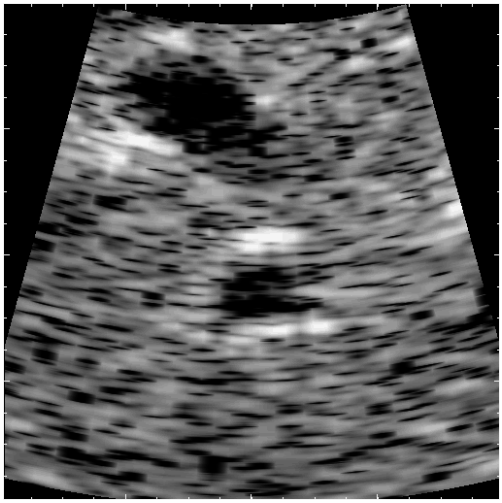


Figure 3.15: $4 \times 4$ cm image of a human liver from a healthy 28-year-old man. The completely dark areas are blood vessels (from [4], Copyright Cambridge University Press).

An enlarged view of an image of a liver is seen in Fig. 3.15. The image has a grainy appearance, and not a homogeneous gray or black level as might be expected from homogeneous liver tissue. This type of pattern is called speckle. The displayed signals are the backscatter from the liver tissue, and are due to connective tissue, cells, and fibrious tissue in the liver. These structures are much smaller than one wavelength of the ultrasound, and the speckle pattern displayed does not directly reveal physical structure. It is rather the constructive and destructive interference of scattered signals from all the small structures. So it is not possible to visualize and diagnose microstructure, but the strength of the signal is an indication of pathology. A strong signal from liver tissue, making a bright image, is, *e.g.*, an indication of a fatty or cirrhotic liver.

As the scattered wave emanates from numerous contributors, it is appropriate to characterize it in statistical terms. The amplitude distribution follows a Gaussian distribution [34], and is, thus, fully characterized by its mean and variance. The mean value is zero since the scattered signal is generated by differences in the tissue from the mean acoustic properties.

Although the backscattered signal is characterized in statistical terms, one should be careful not to interpret the signal as random in the sense that a new set of values is generated for each measurement. The same signal will result, when a transducer is probing the same structure, if the structure is stationary. Even a slight shift in position will yield a backscattered signal correlated with that from the adjacent position. The shift over which the signals are correlated is essentially dependent on the extent of the ultrasound field. This can also be seen from the image in Fig. 3.15, as the laterally elongated white speckles in the image indicate transverse correlation. The extent of these speckle spots is a rough indication of the point spread function of the system.

The correlation between different measurements is what makes it possible to estimate blood velocities with ultrasound. As there is a strong correlation for small movements, it is possible to detect shifts in position by comparing or, more strictly, correlating successive measurements of moving structure, *e.g.*, blood cells.

Since the backscattered signal depends on the constructive and destructive interference of waves from numerous small tissue structures, it is not meaningful to talk about the reflection strength of the individual structures. Rather, it is the deviations within the tissue and the composition of the tissue that determine the strength of the returned signal. The magnitude of the returned signal is, therefore, described in terms of the power of the scattered sig-

nal. Since the small structures reradiate waves in all directions and the scattering structures might be ordered in some direction, the returned power will, in general, be dependent on the relative position between the ultrasound emitter and receiver. Such a medium is called anisotropic, examples of which are muscle and kidney tissue. By comparison, liver tissue is a fairly isotropic scattering medium, when its major vessels are excluded, and so is blood.

It is, thus, important that the simulation approach models the scattering mechanisms in the tissue. This is essentially what the model derived in Chapter 2 does. Here the received signal from the transducer is:

$$p_r(\vec{r},t) = v_{pe}(t) \underset{t}{\star} f_m(\vec{r}) \underset{r}{\star} h_{pe}(\vec{r},t) \tag{3.34}$$

where $\underset{r}{\star}$ denotes spatial convolution. $v_{pe}$ is the pulse-echo impulse, which includes the transducer excitation and the electro-mechanical impulse response during emission and reception of the pulse. $f_m$ accounts for the inhomogeneities in the tissue due to density and propagation velocity perturbations which give rise to the scattered signal. $h_{pe}$ is the pulse-echo spatial impulse response that relates the transducer geometry to the spatial extent of the scattered field. Explicitly written out these terms are:

$$v_{pe}(t) = \frac{\rho}{2c^2} E_m(t) \underset{t}{\star} \frac{\partial^3 v(t)}{\partial t^3}, \qquad f_m(\vec{r}_1) = \frac{\Delta\rho(\vec{r})}{\rho} - \frac{2\Delta c(\vec{r})}{c}, \qquad h_{pe}(\vec{r},t) = h_t(\vec{r},t) * h_r(\vec{r},t) \tag{3.35}$$

So the received response can be calculated by finding the spatial impulse response for the transmitting and receiving transducer and then convolving with the impulse response of the transducer. A single RF line in an image can be calculated by summing the response from a collection of scatterers in which the scattering strength is determined by the density and speed of sound perturbations in the tissue. Homogeneous tissue can thus be made from a collection of randomly placed scatterers with a scattering strength with a Gaussian distribution, where the variance of the distribution is determined by the backscattering cross-section of the particular tissue. This is the approach taken in these notes.

The computer phantoms typically consist of 100,000 or more discrete scatterers, and simulating 50 to 128 RF lines can take several days depending on the computer used. It is therefore beneficial to split the simulation into concurrently run sessions. This can easily be done by first generating the scatterer's position and amplitude and then storing them in a file. This file can then the be used by a number of workstations to find the RF signal for different imaging directions, which are then stored in separate files; one for each RF line. These files are then used to assemble an image. This is the approach used for the simulations shown here in which 3 Pentium Pro 200 MHz PCs can generate one phantom image over night using Matlab 5 and the Field II program.

## 3.6 Synthetic phantoms

The first synthetic phantom consists of a number of point targets placed with a distance of 5 mm starting at 15 mm from the transducer surface. A linear sweep image of the points is then made and the resulting image is compressed to show a 40 dB dynamic range. This phantom is suited for showing the spatial variation of the point spread function for a particular transducer, focusing, and apodization scheme.

Twelve examples using this phantom are shown in Fig. 3.16. The top graphs show imaging without apodization and the bottom graphs show images when a Hanning window is used for apodization in both transmit and receive. A 128 elements transducer with a nominal frequency of 3 MHz was used. The element height was 5 mm, the width was a wavelength and the kerf 0.1 mm. The excitation of the transducer consisted of 2 periods of a 3 MHz sinusoid with a Hanning weighting, and the impulse response of both the emit and receive aperture also was a two cycle, Hanning weighted pulse. In the graphs A – C, 64 of the transducer elements were used for imaging, and the scanning was done by translating the 64 active elements over the aperture and focusing in the proper points. In graph D and E 128 elements were used and the imaging was done solely by moving the focal points.

Graph A uses only a single focal point at 60 mm for both emission and reception. B also uses reception focusing

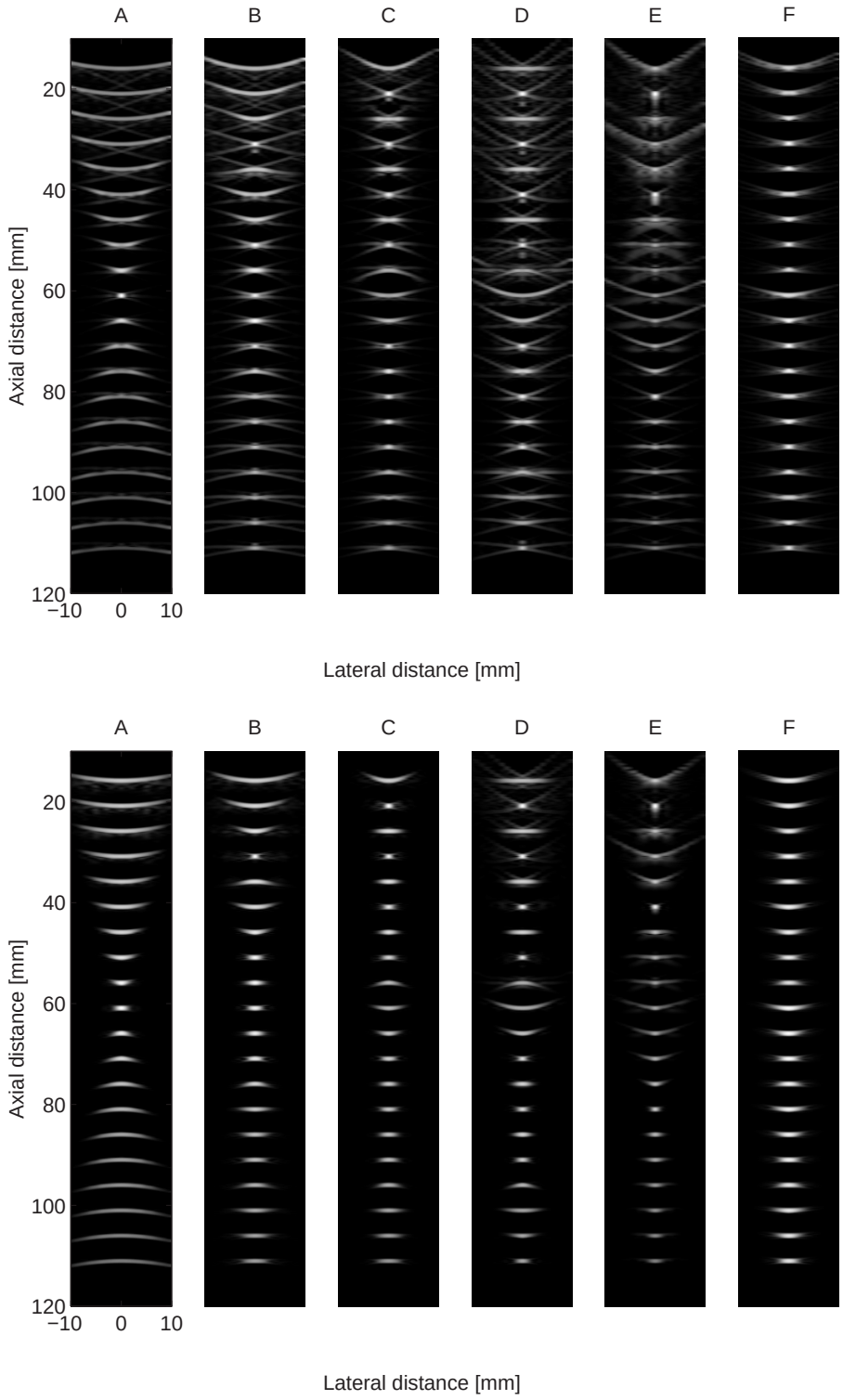



Figure 3.16: Point target phantom imaged for different set-up of transmit and receive focusing and apodization. See text for an explanation of the set-up.

at every 20 mm starting from 30 mm. Graph C further adds emission focusing at 10, 20, 40, and 80 mm. D applies the same focal zones as C, but uses 128 elements in the active aperture.

The focusing scheme used for E and F applies a new receive profile for each 2 mm. For analog beamformers this is a small zone size. For digital beamformers it is a large zone size. Digital beamformer can be programmed for each sample and thus a ”continuous” beamtracking can be obtained. In imaging systems focusing is used to obtain high detail resolution and high contrast resolution preferably constant for all depths. This is not possible, so compromises must be made. As an example figure F shows the result for multiple transmit zones and receive zones, like E, but now a restriction is put on the active aperture. The size of the aperture is controlled to have a constant F-number (depth of focus in tissue divided by width of aperture), 4 for transmit and 2 for receive, by dynamic apodization. This gives a more homogeneous point spread function throughout the full depth. Especially for the apodized version. Still it can be seen that the composite transmit can be improved in order to avoid the increased width of the point spread function at e.g. 40 and 60 mm.

The next phantom consists of a collection of point targets, five cyst regions, and five highly scattering regions. This can be used for characterizing the contrast-lesion detection capabilities of an imaging system. The scatterers in the phantom are generated by finding their random position within a $60 \times 40 \times 15$ mm cube, and then ascribe a Gaussian distributed amplitude to the scatterers. If the scatterer resides within a cyst region, the amplitude is set to zero. Within the highly scattering region the amplitude is multiplied by 10. The point targets has a fixed amplitude of 100, compared to the standard deviation of the Gaussian distributions of 1. A linear scan of the phantom was done with a 192 element transducer, using 64 active elements with a Hanning apodization in transmit and receive. The element height was 5 mm, the width was a wavelength and the kerf 0.05 mm. The pulses where the same as used for the point phantom mentioned above. A single transmit focus was placed at 60 mm, and receive focusing was done at 20 mm intervals from 30 mm from the transducer surface. The resulting image for 100,000 scatterers is shown in Fig. 3.17. A homogeneous speckle pattern is seen along with all the features of the phantom.

## 3.7 Anatomic phantoms

The anatomic phantoms are attempts to generate images as they will be seen from real human subjects. This is done by drawing a bitmap image of scattering strength of the region of interest. This map then determines the factor multiplied onto the scattering amplitude generated from the Gaussian distribution, and models the difference in the density and speed of sound perturbations in the tissue. Simulated boundaries were introduced by making lines in the scatterer map along which the strong scatterers were placed. This is marked by completely white lines shown in the scatterer maps. The model is currently two-dimensional, but can readily be expanded to three dimensions. Currently, the elevation direction is merely made by making a 15 mm thickness for the scatterer positions, which are randomly distributed in the interval.

Two different phantoms have been made; a fetus in the third month of development and a left kidney in a longitudinal scan. For both was used 200,000 scatterers randomly distributed within the phantom, and with a Gaussian distributed scatterer amplitude with a standard deviation determined by the scatterer map. The phantoms were scanned with a 5 MHz 64 element phased array transducer with $\lambda/2$ spacing and Hanning apodization. A single transmit focus 70 mm from the transducer was used, and focusing during reception is at 40 to 140 mm in 10 mm increments. The images consists of 128 lines with 0.7 degrees between lines.

Fig. 3.19 shows the artificial kidney scatterer map on the left and the resulting image on the right. Note especially the bright regions where the boundary of the kidney is orthogonal to the ultrasound, and thus a large signal is received. Note also the fuzziness of the boundary, where they are parallel with the ultrasound beam, which is also seen on actual ultrasound scans. Fig. 3.18 shows the fetus. Note how the anatomy can be clearly seen at the level of detail of the scatterer map. The same boundary features as for the kidney image is also seen.

The images have many of the features from real scan images, but still lack details. This can be ascribed to the low level of details in the bitmap images, and that only a 2D model is used. But the images do show great potential for making powerful fully synthetic phantoms, that can be used for image quality evaluation.

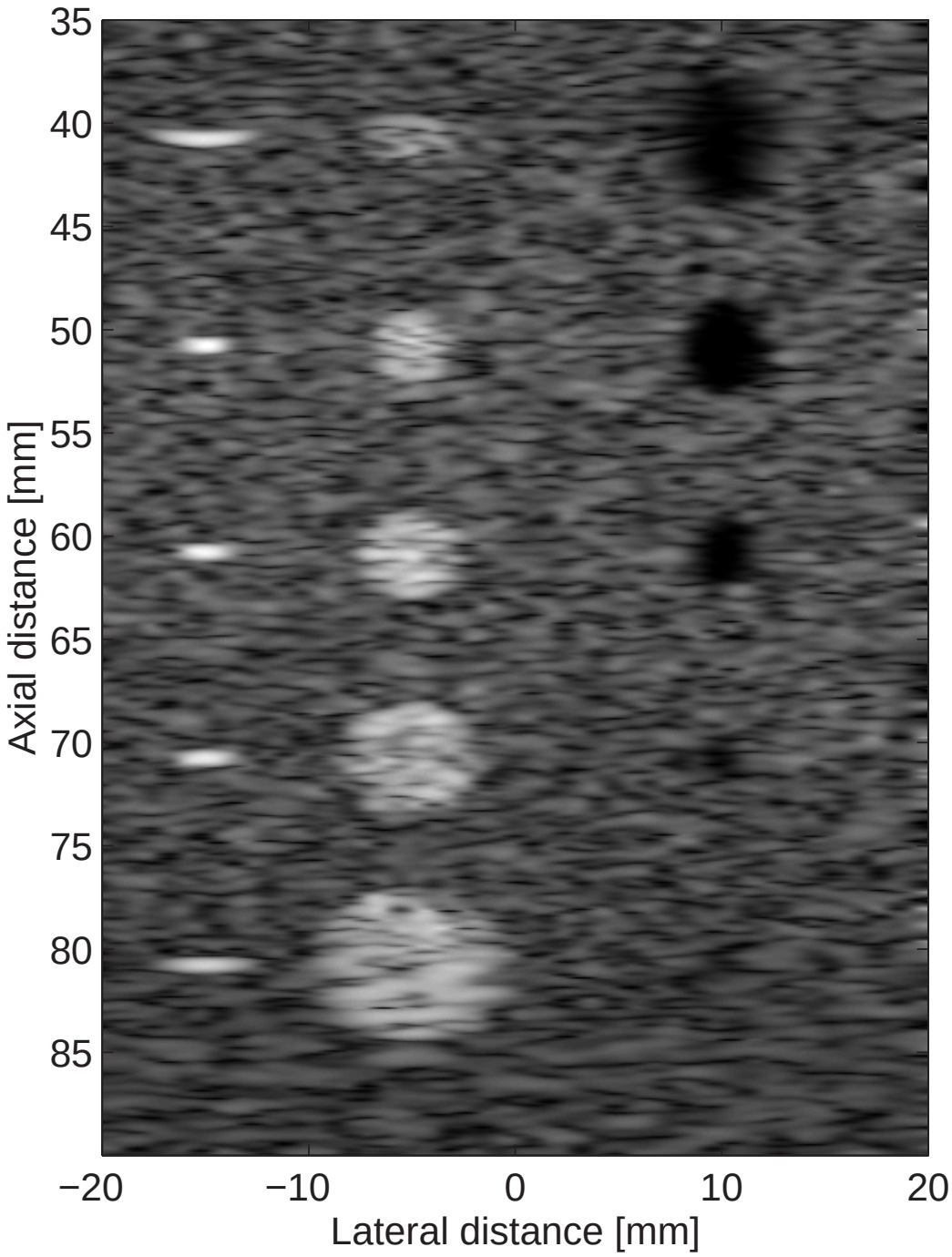



Figure 3.17: Computer phantom with point targets, cyst regions, and strongly reflecting regions.

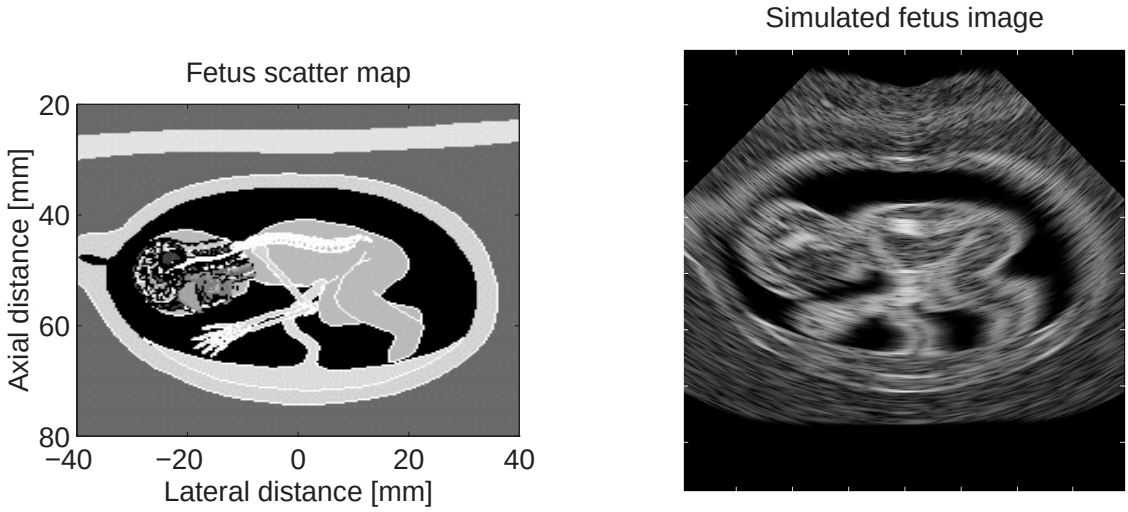



Figure 3.18: Simulation of artificial fetus.

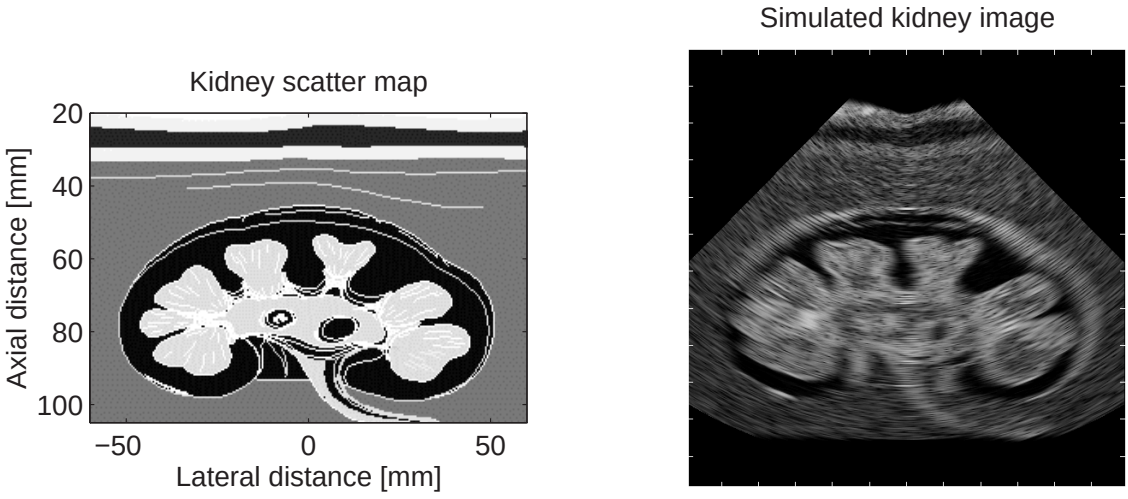



Figure 3.19: Simulation of artificial kidney.

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